



PROFILE

RAJESH AKULA

Institute Post-doctoral Fellow

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EXPERIENCE

POST DOC - 2

2024 (AUGUST) - 2025 (JULY)

University of Toronto, Canada

- Thermal management innovations for enhanced battery performance, longevity, and vehicle range

POST DOC - 1

2023 (MARCH)-2024 (JULY)

Indian Institute of Technology Bombay (IIT Bombay), India

- Smart thermal management of Li-ion cells by predicting the transient velocity profile of the coolant using machine-learning tools to avoid the chances of overcooling or undercooling of the battery pack.

RESEARCH SCHOLAR

2017 (JUL) - 2023 (JAN)

Indian Institute of Technology Madras (IIT Madras), India

Modelling and Thermal management of Li-ion batteries:

- Identified the optimum number of RC circuits and their respective component values in the Equivalent Circuit Model using the Thevenin Equivalent model to predict the Pouch-type battery behavior for various discharge rates.
- Proposed a cost-effective and less intricate thermal management solution for cylindrical Li-ion battery using composite phase change materials.
- Introduced binary alkane mixtures as potential candidates for thermal management of Li-ion batteries and a correlation is developed to quantify their thermal performance.

Thermal management of Electronics:

- Optimized the inter-fin spacing of PCM-based pin fin heat sink for cyclic heat loads using machine learning algorithms.
- Identified the best heat sink configurations when the Integrated Chip (IC) experiences dangerous power surge heat loads.

EDUCATION

M.S AND PH.D.

(DIRECT)

2017 - PRESENT

Indian Institute of Technology Madras, Chennai, India; CGPA - 8.49.

Heat transfer and Thermal Power Laboratory,
Department of Mechanical Engineering.

Thesis title (July 2022): Thermal management of electronics and Li-ion batteries using phase change material-based heat sinks.

BACHELOR OF TECHNOLOGY

2012 - 2016

Rajiv Gandhi University of Knowledge Technologies Nuzvid, India; CGPA - 8.92.

Project title: Effect of heat treatment on microstructure and mechanical properties of AZ91Mg Alloy.

AWARDS AND ACHIEVEMENTS

- Awarded postdoctoral equivalent fellowship by IIT Madras (NIRF - 1) for submitting the final thesis before the maximum period of the Ph.D. program, in the year 2022.
- Secured All India Rank of 1549 out of 190648 candidates in Graduate Aptitude Test in Engineering (GATE) with 99.1875 percentile, organized by Indian Institute of Technology Roorkee, in the year 2017.

TECHNICAL SKILLS

- **Programming languages** : C, Python, MATLAB
- **Software packages** : ANSYS Fluent, COMSOL, SOLIDWORKS, CATIA
- **Machine Learning and optimization Algorithms** : GPR, ANN, GA, NSGA - II

PUBLICATIONS

JOURNAL ARTICLES

- **R. Akula**, C. Balaji, "Thermal management of 18650 li-ion battery using novel fins-PCM-eg composite heat sinks", Applied Energy 316 (2022) 119048 (Impact Factor: 11.446).
<https://doi.org/10.1016/j.apenergy.2022.119048>
- **R. Akula**, A. Minnikanti, C. Balaji, "Pin fin-PCM composite heat sink solution for thermal management of cylindrical Li-ion battery", Applied Thermal Engineering 248 (2024) 123146 (Impact Factor: 6.465).
<https://doi.org/10.1016/j.applthermaleng.2024.123146>
- **R. Akula**, A. Gopinath, S. Rangarajan, C. Balaji, "Experimental and numerical studies on heat transfer from a PCM-based heat sink with baffles", International Journal of Thermal Sciences 159 (2020) 106525 (Impact Factor: 4.779).
<https://doi.org/10.1016/j.ijthermalsci.2020.106525>
- **R. Akula**, C. Balaji, "Effect of volume fraction of fins on the thermal management for various cyclic heat loads appear in electronics", Heat Transfer Engineering 1-21 (2023) (Impact Factor: 2.431).
<https://doi.org/10.1080/01457632.2023.2295083>
- **R. Akula**, C. Balaji, "Thermal Performance of a phase change material-based heat sink subject to constant and power surge heat loads: A numerical study", Journal of Thermal Science and Engineering Applications 13 (2020) (Impact Factor: 1.47).
<https://doi.org/10.1115/1.4047944>
- S. Kumar, **R. Akula**, C. Balaji, "An inverse methodology to estimate the thermal properties and heat generation of a Li-ion battery", Applied Thermal Engineering 236 (2024) 121752 (Impact Factor: 6.465).
<https://doi.org/10.1016/j.applthermaleng.2023.121752>
- S. Thakur, **R. Akula**, L. Kumar, "An Effective and Lightweight battery thermal management system with incremental contact area: A numerical study", Applied Thermal Engineering 242 (2024) 122555 (Impact Factor: 6.465).
<https://doi.org/10.1016/j.applthermaleng.2024.122555>
- K. Dhuper, **R. Akula**, S. Duttagupta, L. Kumar, "Enhancement of heat transfer using conical fins for electronic cooling and their design optimization for maximum thermal regulation using ANN assisted NSGA II", International Journal of Thermal Sciences 202 (2024) 109092 (Impact Factor: 4.779)
<https://doi.org/10.1016/j.ijthermalsci.2024.109092>

- E. Shrinet, **R. Akula**, L. Kumar, “Improvement in serpentine channel based battery thermal management system geometry using variable contact area and its multi-objective design optimization”, Journal of Energy Storage 96 (2024) 112726 (Impact Factor: 8.9) <https://doi.org/10.1016/j.est.2024.112726>
- S. Mujumdar, **R. Akula**, C. Balaji, “Thermal management of electronics under cyclic heat loads using graphite-assisted heat sink”, Heat Transfer Engineering 46 (2024) (Impact Factor: 2.431) - In Press.

CONFERENCE PROCEEDINGS

- **R. Akula**, L. Kumar, “Prediction of the minimum heat loss coefficient for safe operation of a Li-ion cell: A machine learning approach”, The 9th European Thermal Sciences Conference (Eurotherm 2024), June 10-13, 2024, Bled, Slovenia.
- **R. Akula**, C. Balaji, Effect of PCM fill ratio and heat sink orientation on the thermal management of transient power spikes in electronics, Journal of Physics: Conference Series, 2116 (2021) 012077 (Impact Factor: 0.547).
- **R. Akula**, C. Balaji, “Optimization of dimensions and fin volume fractions of PCM based heat sinks for cyclic heat load”, The 8th Asian Symposium on Computational Heat Transfer and Fluid Flow, Sep. 23-26, 2021, Qingdao, China.
- **R. Akula**, C. Balaji, “Characterization of PCM based heat sinks for various power surge durations and intervals: An experimental study”, Proceeding of the 26th National and 4th International ISHMT-ASTFE Heat and Mass Transfer Conference (IHMTTC-2021), December 17-20, 2021, IIT Madras, Chennai, India.
- **R. Akula**, S. Rangarajan, C. Balaji, “Thermal Management of Transient Power Spikes in Electronics using a solid-liquid Phase Change Material based heat sinks: A numerical study”, Proceeding of the 25th National and 3rd International ISHMT-ASTFE Heat and Mass Transfer Conference (IHMTTC-2019), December 28-31, 2019, IIT Roorkee, Roorkee, India.
- S. Majumdar, **R. Akula**, C. Balaji, “Design of thermal management system for a battery pack with malfunctioning battery: - a numerical parametric study”, The 9th European Thermal Sciences Conference (Eurotherm 2024), June 10-13, 2024, Bled, Slovenia.
- S. Majumdar, **R. Akula**, C. Balaji, “Thermal management of electronics working on cyclic heat loads: An experimental study”, Proceedings of the 17th International Heat Transfer Conference, IHTC-17, 14 – 18 August 2023, Cape Town, South Africa.
- K. Vasista, **R. Akula**, C. Balaji, “Thermal management of electronic chip using minichannel based hybrid cooling system for power surge heat loads”, Proceedings of the 17th International Heat Transfer Conference, IHTC-17, 14 – 18 August 2023, Cape Town, South Africa.

PRESENTATIONS AND INVITED TALKS

- Delivered an invited talk titled "Thermal management of Li-ion batteries" in Mechanics workshop, IIT Madras, India, April 2022.
- Virtually presented the work titled "Characterization of PCM-based heat sinks for various power surge durations and intervals: An experimental study" in ISHMT, IIT Madras, Chennai, December 2021.

- Virtually presented the work titled "Optimization of dimensions and fin volume fractions of PCM-based heat sinks for cyclic heat load" in ASCHT, China, September 2021.
- Presented the work titled "Thermal Management of Transient Power Spikes in Electronics using a solid-liquid Phase Change Material based heat sinks: A numerical study" in IHMTC, Roorkee, India, December 2019.

MENTORSHIP

- Abhishek R (Undergraduate student 2020, IIT Madras) - Performance of PCM-based heat sinks under cyclic heat loading witnessed by electronics.
- Abhiram M (Post Graduate student 2021, IIT Madras) - PCM based heat sink solution for optimal battery thermal management
- Nityan S (Post Graduate student 2022, IIT Madras) - Thermal modeling and design optimization of mini channel cold plate for Li-ion battery pack.
- Pratyusha A (Post Graduate student 2022, IIT Madras) - Thermal characterization of Li-ion batteries for various discharge rates using the NTGK model.
- Ameya A (Post Graduate student 2023, IIT Madras) - Heat transfer correlations for PCM-based thermal management of Li-ion batteries.
- Zyed B (Post Graduate student 2023, IIT Madras) - Design of PCM-based heat sink for thermal management of Li-ion batteries using topology optimization.
- Sandeep A (Post Graduate student 2023, IIT Madras) - Thermal characterization of cold plate with heat pipe and PCM for effective cooling of Li-ion batteries.

ASSISTANTSHIPS

TEACHING ASSISTANT

IIT Madras, Chennai

- ME5110: Inverse Methods in heat transfer - Graduate level (Winter 2019, 2020, 2021)
- ME3103: Energy conversion system - Undergraduate level (Summer 2019)
- Heat Transfer Engineering: Fundamentals and Techniques (2021), C. Balaji, Balaji Srinivasan, and Sateesh Gedupudi. Academic Press, ISBN: 9780128185032
- Essentials of Radiation Heat Transfer (2021), C. Balaji. Springer, Cham ISBN: 9783030626167
- Thermal System Design and Optimization 2nd Edition (2019), C. Balaji. Ane Books Pvt. Ltd. New Delhi ISBN: 9789388264785.
- Part of the organizing team for the 6th Asian Symposium on Computational Heat Transfer and Fluid Flow, organized by the Indian Institute of Technology Madras, Chennai, from 10 to 13 December 2017.
- Volunteered for conducting M.S./Ph.D. admission written tests (IIT Madras).

BOOK ASSISTANSHIP

OTHERS

REFERENCES

Prof. C Balaji (Ph.D. supervisor)

TT Narendran Chair Professor,
Department of Mechanical Engineering,
IIT Madras, Chennai, India.
E-mail: balaji@iitm.ac.in

Dr. Lalit Kumar (Postdoc - 1 supervisor)

Department of Energy Science and
Engineering,
IIT Bombay, Mumbai, India.
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Prof. Cristina Amon (Postdoc - 2 supervisor)

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University of Toronto, Canada.
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